Fiscal Year:	FY 2004	**Task Last Updated:**	FY 08/28/2004
PI Name:	Parthasarathy, Ramkumar Ph.D.		
Project Title:	Instability and Breakup of Gas Jets Injected in Coflowing Liquids		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Dynamics and stability		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	73019	**Congressional District:**	4
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	04/01/2000	**End Date:**	11/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	4
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	Note - date changed from 11/30/2003		
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Gollahalli, S R (University of Oklahoma)		
Grant/Contract No.:	NCC3-787		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary focus of this effort is on gas jet injection into a co-flowing liquid medium over a range of jet diameters, phase velocities, and fluid properties including viscosities and densities. There are three tasks: Testing in normal gravity, testing in microgravity, and theoretical model development. Both testing phases will involve design, fabrication and checkout of experimental hardware. The microgravity testing will occur in the 2.2-Second Drop Tower and, possibly, the KC-135 aircraft. Model development will focus on determining the absolute and convective instability ranges of the jet, developing a non-linear theory of gas jet breakup by varicose disturbances and comparison of the model with experimental data. This effort will delineate the various regimes of instabilities that occur in gas jets injected into a liquid medium and then isolate and quantify the effects of buoyancy on the instabilities leading to the jet breakup. The breakup of gas jets injected into liquids and the resulting bubble formation are used in number of practical applications, such as gas dissolution in liquids used gas-liquid reactors and blood oxygenators, coal and mineral purification by		

	flotation, and large-scale mixing in liquids. Inherent instabilities in the flow get amplified and result in the breakup of the jet. Thus, important properties, e.g., the breakup distance, the sizes of the bubbles formed as a result of breakup, are determined by the nature of instabilities that lead to the breakup.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	The effects of gravity on the bubble formation in an annular jet were studied. The experiments were conducted in the 2.2-second drop tower at the NASA Glenn Research Center. Terrestrial gravity experiments were conducted at the Fluid Dynamics Research Laboratory at the University of Oklahoma. Stainless steel tubing with inner diameters of 1/8" (gas inner annulus) and 5/16" (liquid outer annulus) served as the injector. A rectangular test section, 6" x 6" x 14" tall, made out of half-inch thick Lexan was used. Images of the annular jet were acquired using a high-speed camera. The effects of gravity and varying liquid and gas flow rates on bubble size, wavelength, and breakup length were documented. In general, the bubble diameter was found to be larger in terrestrial gravity than in microgravity for varying Weber numbers (0.05 – 0.16 and 5 – 11) and liquid flow rates (1.5 ft/s – 3.0 ft/s). The wavelength was found to be larger in terrestrial gravity than in microgravity, but remained constant for varying Weber numbers. For low Weber numbers (0.05 – 0.16), the breakup length in microgravity was significantly higher than in terrestrial gravity. The significant conclusions drawn from these experiments were: 1. The bubble size was larger in terrestrial conditions than in microgravity. This could be due to the result of buoyancy of the gas annulus, the reduced relative velocities of the two phases due to gravity, and the possible entrainment of gas into the liquid under terrestrial conditions. 2. Bubble formation frequency appeared to be independent of gas velocity in terrestrial gravity but was highly dependent upon gas velocity in reduced gravity conditions. 3. The wavelength was larger in terrestrial gravity, which would stretch the jet. The wavelength did not appear to be a direct function of liquid velocity and may be more controlled by gas velocities. 4. The breakup length was significantly larger in microgravity. The accelerated liquid in terrestrial gravity resulted in the jet pinching together and separating. For high Weber numbers, the breakup length was too large to be viewed.
Bibliography Type:	Description: (Last Updated: 06/09/2004)
Abstracts for Journals and Proceedings	Koepp, R.; Parthasarathy, R. N. and Gollahalli, S. R. "Effects of Gravity on Bubble Formation in an Annular Jet" 41st AIAA Aerospace Sciences Meeting and Exhibit, Reno AIAA 2004-0965 , Jan-2004
Abstracts for Journals and Proceedings	Parthasarathy, R. N.; Carrera, J.; Koepp, R.; Gollahalli, S. R. "Gas Bubble Formation From a Freestanding Tube in Microgravity" Sixth ISHMT/ASME Heat and Mass Transfer Conference, Kalpakkam, India ISHMT-03 , Jan-2004
Dissertations and Theses	Koepp, R. "Effects of Gravity on Bubble Formation in an Annular Jet" School of Aerospace and Mechanical Engineering, The University of Oklahoma, Norman, Oklahoma Aug-2003